



Addis Ababa University
Department of Computer Science
Semester 1, Year 2025/2026
Project Chapter 1 -2

Project Title	AidHarmony
---------------	------------

Submitted by:

ID	Student Name
NSE/8499/14	Yordanos Getaneh
NSE/8924/14	Yonathan Degsera
NSE/0878/14	Abemelek behailu

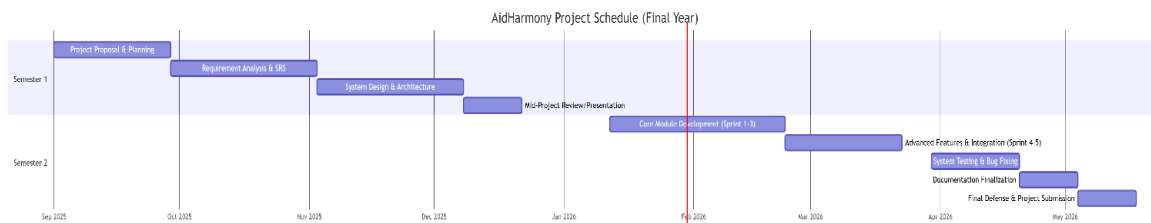
Submitted to:

Mrs Fikirte G.

2/3/2025
Addis Ababa
Ethiopia

Contents

Chapter 1	1
Project Proposal	1
1.1 Introduction	1
1.2 Statement of the Problem and Justification	1
Justification for the New Focus:	1
1.3 Project Objective	2
1.3.1 General Objective of the System	2
1.3.2 Specific Objective of the System	2
1.4 Scope of the Project	2
1.5 System Development Methodology	3
1.5.1 Investigation (Fact-Finding) Methods	3
1.5.2 System Development Tools	3
1.6 Significance of Project	4
1.6.1 Societal Significance	4
1.6.2 Technological Significance	5
1.7 Beneficiaries	5
1.8 Time Schedule	5



Chapter 2	6
Requirement Analysis	6
2.1 Introduction	6
2.2 Current System	6
2.2.1 Major Function of the Current System / Current System Description	6
2.2.2 Problem of the Existing System	6
2.3 Requirement Gathering	7
2.3.1 Requirement Gathering Methodologies	7
2.3.2 Results Found	7
2.4 Proposed System	7
2.4.1 Overview	7

2.4.2 Functional Requirements	8
2.4.3 Non-Functional Requirements.....	9
2.5 Constraints / Pseudo Requirements	10
2.6.1 System Model	11
2.6.2 Use Case Model	12
2.6.2.1 Use Case Diagram	15
2.6.2.2 Description of Use-Case Model.....	18
2.6.3.2 Class Modeling	26
2.7 Dynamic model	27
2.7.1Sequence Diagram	27
2.7.2 Flow diagrams.....	30

Chapter 1

Project Proposal

1.1 Introduction

In the digital age, charitable giving in Ethiopia faces a significant "trust gap." While global platforms exist, they often focus exclusively on monetary transactions and lack the localized transparency needed to reassure donors. AidHarmony is a dedicated web platform designed to empower registered charitable organizations by providing them with a robust infrastructure to manage diverse campaigns.

The platform's core innovation is no longer individual vetting, but rather the provision of a professional suite for organizations to demonstrate impact through milestone-based reporting, digital receipts, and multi-modal support (money, materials, and time). By focusing on established entities, AidHarmony shifts from being a verification authority to a transparency enabler.

1.2 Statement of the Problem and Justification

Donors are often hesitant to contribute due to concerns about fund mismanagement and a lack of visible outcomes. Existing systems treat charity as a one-time transaction, offering no structured way to track progress or donate non-financial resources like medical supplies or professional skills.

Justification for the New Focus:

- **Feasibility:** By serving established charity companies, the system bypasses the legal and operational complexity of verifying private individuals, focusing instead on organizations with existing legal standing.
- **Trust through Transparency:** The system solves the "trust deficit" by mandating that organizations provide campaign reports and milestone updates to unlock further engagement.
- **Holistic Support:** It provides a unique advantage for companies that need materials (donations-in-kind) or volunteers, which are often as vital as financial aid.

1.3 Project Objective

1.3.1 General Objective of the System

To design and develop AidHarmony, a transparent web-based management system that enables charitable organizations to run multi-modal campaigns, provide verifiable impact reports, and issue digital receipts to secure donor trust.

1.3.2 Specific Objective of the System

1. To implement a professional dashboard for Charitable Organizations to manage complex aid campaigns.
2. To develop a Milestone & Reporting Engine that allows organizations to upload evidence of impact (photos, documents, receipts).
3. To integrate a Digital Receipting System that provides donors with immediate proof of their financial or material contributions.
4. To enable Multi-Modal Giving, allowing donors to pledge specific material items or volunteer hours directly to organizational projects.
5. To provide an Administrative Audit Trail to monitor organizational compliance and transparency.

1.4 Scope of the Project

This project covers the full software development lifecycle for the AidHarmony web platform, from requirement analysis and system design to implementation, testing, and deployment.

What the project includes (In-Scope):

- **Organizational Portal:** Tools for charity companies to create and track campaigns.
- **Transparency Module:** Milestone trackers and public impact reporting features.
- **Donation Engine:** Processing for cash, material pledges, and volunteer applications.
- **Receipting System:** Automated generation of digital donation receipts.

What the project excludes (Out-of-Scope):

- Development of native mobile applications (iOS/Android). The platform will be a responsive website accessible via mobile browsers.
- Physical logistics management for the delivery of material goods. The platform will facilitate the pledge and connection but not handle shipping or transportation.
- Legal and financial advisory services to users.
- Direct banking operations or acting as a financial intermediary. All funds are processed through third-party payment gateways.
- Advanced Artificial Intelligence (AI) modules for automated fraud detection or document analysis in the initial version.

1.5 System Development Methodology

1.5.1 Investigation (Fact-Finding) Methods

The requirements for AidHarmony were gathered through a combination of methods to ensure a comprehensive understanding of the problem domain and user needs:

- **Analysis of Existing Systems:** A critical study of platforms like GoGetFunding and GoFundMe was conducted to identify their strengths and common shortcomings regarding verification and transparency.
- **Literature Review:** Academic and industry literature on crowdfunding challenges, particularly in developing economies like Ethiopia, was reviewed to understand contextual barriers such as trust and payment integration.
- **Stakeholder Analysis:** Potential user roles (Donor, Volunteer, Beneficiary, Admin) were analyzed to derive their core needs, pain points, and expected interactions with the system.

1.5.2 System Development Tools

The project will adhere to an **Agile Software Development methodology**, specifically using the **Scrum framework**. This iterative and incremental approach is chosen for its flexibility, focus on delivering working software, and ability to incorporate feedback regularly.

- **Project Management:** Scrum roles (Product Owner, Scrum Master, Development Team), Sprints (2-3 week iterations), and artifacts (Product Backlog, Sprint Backlog) will be used.
- **Development Tools:** The technology stack includes:
 - **Backend:** Python with Django web framework and Django REST Framework (DRF) for APIs.
 - **Frontend:** JavaScript with React library for building the user interface.
 - **Database:** PostgreSQL for persistent data storage.
 - **Version Control:** Git with GitHub for source code management and collaboration.
 - **Other Tools:** Docker for containerization, and various testing libraries (e.g., Pytest).

1.6 Significance of Project

The AidHarmony project holds substantial societal and technological significance by addressing the fundamental barrier to digital philanthropy: the trust deficit between donors and aid recipients.

1.6.1 Societal Significance

- **Empowering Institutional Giving:** By providing a specialized web system for registered charity companies, the platform strengthens the capacity of the non-profit sector to mobilize resources more efficiently than through fragmented individual efforts.
- **Restoring Public Trust:** The system's focus on milestone reporting and campaign progress updates ensures that donors are not just "giving and forgetting," but are active participants in a transparent aid lifecycle.
- **Guaranteeing Donor Confidence:** The integration of a digital receipting system and mandatory impact evidence (documents/photos) provides donors with a sense of security and a tangible "guarantee" that their contributions—whether financial or material—have reached the intended destination.
- **Promoting Holistic Support:** Unlike traditional cash-only platforms, AidHarmony highlights the importance of non-cash support (donations-in-kind and volunteering), allowing a broader segment of the community to contribute their skills and goods to meaningful causes.

1.6.2 Technological Significance

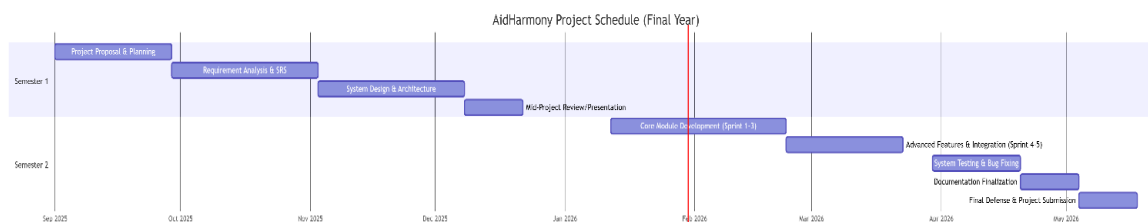
- **Practical Full-Stack Implementation:** The project demonstrates the application of modern web principles using the Django-React stack to solve complex real-world problems like secure payment processing and automated receipt generation.
- **Advanced Data Modeling:** It tackles the challenge of designing a multi-modal contribution system where different types of aid (money, items, time) are tracked and reported under a single campaign architecture.
- **Accountability-Driven Design:** The project serves as a model for "Transparency by Design," implementing features like **public transaction logs** and **standardized utilization reports** that are essential for modern financial and charitable software.

1.7 Beneficiaries

- **Charitable Organizations:** Gain a professional web system to reach donors and prove their legitimacy through data.
- **Donors:** Receive a "guaranteed" feeling through transparent progress reports and official receipts.
- **Volunteers:** Access structured service opportunities within established organizations

1.8 Time Schedule

The project will be executed over the final academic year, divided into two semesters. The following Gantt chart outlines the high-level schedule for major phases and deliverables:



Chapter 2

Requirement Analysis

2.1 Introduction

The purpose of this chapter is to provide a structured specification for the AidHarmony system. It transitions from an analysis of the existing landscape to a detailed model of a solution designed specifically for registered charitable organizations. The chapter establishes the functional and non-functional requirements necessary to build a platform where trust is not just promised, but is technically guaranteed through structured reporting and verifiable milestones.

2.2 Current System

2.2.1 Major Function of the Current System / Current System Description

Primary Actors and Functions in Existing Systems

Actor	Primary Function	Interaction with System
Charitable Organization	Manages high-impact, multi-modal campaigns (money, materials, skills).	Sets campaign milestones, uploads impact reports, and manages material/volunteer needs.
Donor / Volunteer	Provides diverse support (cash, goods, or expertise) with a need for verification.	Makes contributions, tracks progress via milestones, and receives automated digital receipts.
AidHarmony System	Acts as a transparency and trust-enabling engine.	Generates legal receipts, tracks milestone progress, and mandates reporting for campaign closure.

2.2.2 Problem of the Existing System

- **The Individual Trust Barrier:** Current systems are plagued by fraudulent individual campaigns because verifying thousands of private citizens is logistically impossible for a platform.
- **Monolithic Aid:** They focus almost exclusively on cash, ignoring that organizations often desperately need material goods (medical supplies, food) and skilled labor (volunteers).
- **Lack of Closure & Receipts:** Donors rarely receive formal proof of how their money was spent or a "guaranteed" receipt that can be used for their own records or tax purposes.

2.3 Requirement Gathering

2.3.1 Requirement Gathering Methodologies

The requirements for the AidHarmony system were derived using a multi-faceted approach:

- **Analysis of Existing Systems:** A critical feature and gap analysis of leading platforms like GoGetFunding was conducted to understand standard functionalities and identify their limitations regarding verification, transparency, and contribution types.
- **Literature Review:** Scholarly articles and industry reports on crowdfunding, particularly in developing contexts like Ethiopia, were reviewed. This highlighted contextual challenges such as digital payment barriers, the importance of social trust, and the need for localized solutions.
- **Stakeholder and Role Analysis:** Potential users of the system were categorized into primary roles (Donor, Volunteer, Beneficiary, and Administrator). The needs, goals, and pain points for each role were analyzed to derive actor-centric requirements and user stories.

2.3.2 Results Found

The investigation confirmed a strong market need for a platform that prioritizes **trust through verification** and **impact through transparency**. Key findings include:

1. **Trust is the Primary Barrier:** Donors' willingness to give is directly tied to their confidence in the beneficiary's legitimacy and the platform's ability to ensure proper fund use.
2. **Needs are Multidimensional:** Community projects often require a mix of money, specific goods, and skilled labor. A platform facilitating only one type limits its effectiveness.
3. **Transparency is Non-Negotiable:** Donors expect and deserve clear, post-donation communication about how their contribution made a difference.
4. **Local Context Matters:** Features like local language support, integration with familiar payment methods, and geolocation are crucial for adoption in specific regions.

2.4 Proposed System

2.4.1 Overview

AidHarmony is a next-generation management system for **Charitable Organizations**. Its core proposition is to move from "Blind Giving" to "**Guaranteed Impact.**" The system provides organizations with tools to manage multi-modal campaigns (Money, Materials, and Volunteers)

and mandates **Milestone-Based Reporting** to maintain donor trust.

2.4.2 Functional Requirements

1. The system shall allow charitable organizations to create and manage professional profiles including mission statements and legal registration details.
2. The system shall enable organizations to initiate multi-modal campaigns with specific targets for financial aid, material goods, and volunteer expertise.
3. The system shall allow organizations to define a roadmap for each campaign by setting specific, measurable milestones.
4. The system shall provide a dedicated portal for organizations to upload "Impact Evidence," such as site photos, delivery receipts, and progress reports for each milestone.
5. The system shall automatically generate a secure, downloadable PDF digital receipt for every successful donation.
6. The system shall include a unique transaction hashing or ID system on every receipt to ensure the "guaranteed" feeling for donors.
7. The system shall provide donors with a personalized dashboard to track the specific progress of the campaigns they have supported.
8. The system shall allow donors to pledge specific items from a list of required materials provided by the organization.
9. The system shall feature a "Skill-Matching" algorithm that connects volunteers to organizational tasks based on professional tags (e.g., Accounting, Medical, Teaching).
10. The system shall maintain a real-time progress bar for each campaign reflecting the percentage of cash, materials, and volunteers secured.
11. The system shall restrict the start of new campaign phases until the organization has uploaded reports for the previous milestone.
12. The system shall send automated email and push notifications to supporters whenever an organization posts a milestone update or an impact report.
13. The system shall provide a public-facing transparency ledger where any user

can view the history of resource utilization by a specific organization.

14. The system shall allow volunteers to log their hours, which the organization must then verify to contribute to the campaign's success metrics.
15. The system shall feature an administrative dashboard to monitor organizational activity and flag profiles that fail to provide required milestone evidence.
16. The system shall integrate a geolocation-based search to help donors and volunteers find organizational needs in their immediate vicinity.
17. The system shall provide an "End-of-Campaign" summary report automatically generated for all contributors once all milestones are met.
18. The system shall allow organizations to export donor lists and contribution data for their own internal auditing and tax reporting.

2.4.3 Non-Functional Requirements

User Interface and Human Factors

- The platform shall achieve a System Usability Scale (SUS) score of at least 80.
- The interface shall be fully responsive (mobile & desktop) and adhere to WCAG 2.1 AA accessibility guidelines.
- Critical user-facing text shall be available in both English and Amharic.

Documentation

- Comprehensive technical documentation for developers (API specs, architecture) shall be provided.
- Detailed user manuals for Donors, Volunteers, and Beneficiaries shall be created.

Hardware Considerations

- The system will be deployed on cloud infrastructure (e.g., AWS, DigitalOcean) with no specific client-side hardware requirements beyond a modern web browser.

Performance Characteristics

- The campaign browse page shall load within 2 seconds under a load of 500 concurrent users.
- The payment processing pipeline shall complete 99% of transactions within 10 seconds.

Error Handling and Extreme Conditions

- All user inputs shall be validated, with clear, actionable error messages.

- The system shall log all errors and maintain core functionality (e.g., browsing) even if secondary services (e.g., email notifications) are temporarily unavailable.

Quality Issues

- The core application shall maintain 99.5% monthly uptime.
- The code base shall maintain a test coverage of at least 80% for critical backend modules.

System Modifications

- The system shall be built using a modular, service-oriented architecture to allow for easy addition of new features (e.g., SMS notifications, advanced analytics) in the future.

Physical Environment

- The system is designed for standard data center/cloud environments. It is a web application with no specific physical environmental requirements for end-users.

Security Issues

- All user passwords shall be hashed using bcrypt. All data transmission shall use TLS 1.2+.
- A strict Role-Based Access Control (RBAC) model shall be implemented.
- All sensitive actions (logins, financial transactions, admin changes) shall be logged in an immutable audit trail.

Resource Issues

- The system shall be containerized (Docker) to allow efficient horizontal scaling of stateless components based on load.

2.5 Constraints / Pseudo Requirements

1. **Legal and Compliance Constraints:** The system must be designed to comply with relevant data protection regulations (e.g., GDPR principles) and local financial transaction guidelines in Ethiopia.
2. **Technological Constraints:** The initial implementation will be a responsive web application, not a native mobile app, due to scope and resource limitations.
3. **Third-Party Dependency:** Core functionalities like payment processing are dependent on the stability and APIs of third-party gateway services (e.g., telebirr, Chapa).
4. **Operational Constraint:** The verification process requires active, manual review by an administrator, which may create a bottleneck during periods of high registration volume.

2.6.1 System Model

Scenarios

Scenario ID	Scenario Name	Primary Actor	Description & Flow
SC-01	Organization Profile Setup	Charity Organization	The organization registers using official NGO/Company credentials. They set up a professional profile that serves as the "Trust Center" for their future campaigns.
SC-02	Multi-Modal Campaign Launch	Charity Organization	The representative creates a campaign with three specific goal types: Financial (Cash), Material (Physical Goods), and Volunteer (Skilled Labor). This allows for holistic aid.
SC-03	Financial Donation & Auto-Receipting	Donor	A donor contributes money. The system instantly generates a secure PDF receipt with a unique Hash ID, providing an immediate guarantee to the donor.
SC-04	Milestone Achievement & Evidence Upload	Charity Organization	Upon reaching a set goal (e.g., "Buying Desks"), the organization must upload photos/invoices. The system locks the next phase until this transparency requirement is met.
SC-05	Material Pledging & Handover	Donor	A donor pledges "10 Computers." The system provides a pledge voucher. Once the organization confirms physical receipt, the material progress bar updates automatically.
SC-06	Skill-Based Volunteer Matching	Volunteer	A professional (e.g., an Engineer) finds a task matching their skills. They apply, and the organization manages their contribution through the platform's task tracker.
SC-07	Evidence Rejection & Correction	Admin / System	If an organization uploads blurry or invalid evidence for a milestone, the system/admin flags it. The campaign is paused until valid proof of impact is provided.
SC-08	Impact Progress Notification	System	Whenever a milestone is verified, the system automatically blasts an "Impact Update" email to every donor who contributed, closing the feedback loop and building long-term trust.

Scenario ID	Scenario Name	Primary Actor	Description & Flow
SC-09	Campaign Audit & Final Closure	System Administrator	Once all goals are met, the admin reviews the full audit trail of receipts and milestone photos before marking the campaign as "Successfully Closed" for public record.
SC-10	Donor Tax/History Export	Donor	At the end of the year, a donor can export a consolidated report of all their digital receipts for tax purposes or personal records of their social impact.

2.6.2 Use Case Model

Actor Name	Type	Description & Responsibilities
Charity Organization Representative	Primary / Beneficiary	The authorized user representing a registered NGO or charitable company. They are the primary beneficiaries of the system's tools. Their responsibilities include setting up the organization's profile, launching multi-modal campaigns, defining milestones, and uploading evidence/reports to maintain transparency.
Donor	Secondary / Supporter	An individual or corporate entity that provides financial or material support. They use the system to discover verified campaigns, track their contributions through the milestone dashboard, and receive automated digital receipts as a guarantee of their donation.
Volunteer	Secondary / Supporter	A skilled individual looking to contribute time and expertise. They browse campaigns for specific "Volunteer Roles" (e.g., medical, teaching, or technical tasks), apply for positions, and log their service hours for organizational verification.

Actor Name	Type	Description & Responsibilities
System Administrator	Administrative	A high-level user responsible for the platform's integrity. They do not verify the legal status of individuals but monitor organizational compliance, review flagged milestone reports, manage the system's "Trust Score" algorithm, and provide technical support to all users.
Payment Gateway (External)	System / Actor	An external API (e.g., Chapa, Telebirr, or PayPal) that interacts with the system to process financial transactions securely. It sends confirmation signals to the system to trigger the automatic receipting engine.

Detailed Actor Goal Analysis

To ensure this project is "life-changing" and highly professional, we categorize what each actor specifically seeks to achieve through the system:

1. Charity Organization Representative (The "Power User")

- Goal: To professionalize their resource mobilization.
- Key Actions: * Creating campaigns that ask for more than just money (Materials + Volunteers).
 - Using the Milestone Engine to build a history of "Proven Impact" which attracts more donors.
 - Exporting donor data and receipt history for their own legal auditing.

2. Donor (The "Guaranteed Giver")

- Goal: To give without fear of fraud.
- Key Actions: * Receiving an Immediate PDF Receipt with a unique transaction hash.
 - Monitoring the Milestone Progress Bar to see exactly which phase of the project their money is currently funding.
 - Viewing uploaded photos/documents provided by the organization as "Proof of

Life" for the project.

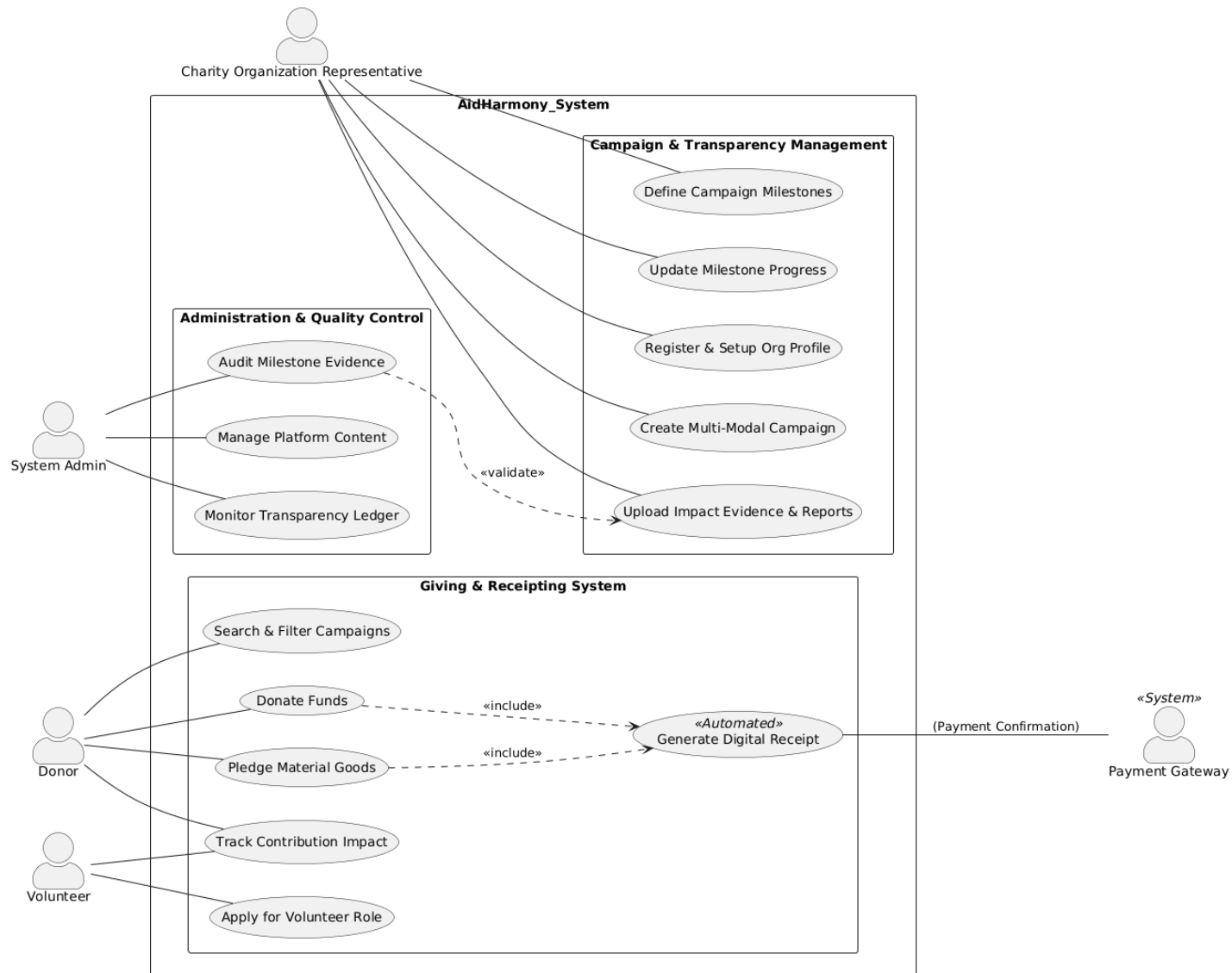
3. Volunteer (The "Skilled Contributor")

- Goal: To find meaningful, verified service opportunities.
- Key Actions: * Filtering campaigns by Skill Tags to ensure their time is used effectively.
 - Receiving a "Service Certificate" or verified log of hours once the organization confirms their task completion.

4. System Administrator (The "Auditor")

- Goal: To maintain platform quality and trust.
- Key Actions: * Auditing campaigns that have high donor complaints or missing milestone evidence.
 - Managing the global database of digital receipts to prevent duplication or fraud.

2.6.2.1 Use Case Diagram



Mapping of FR's to use cases

Use Case ID	Use Case Name	Related FR	Description
UC1	Register & Manage Org Profile	FR1	Allows charity companies to establish their legal identity. The system stores mission statements and legal registration IDs to ensure institutional

Use Case ID	Use Case Name	Related FR	Description
			accountability.
UC2	Create Multi-Modal Campaign	FR2, FR3	Included: UC3. The organization defines targets for cash, materials, and skills. It forces the creation of a roadmap (milestones) before the campaign goes live.
UC3	Set Campaign Milestones	FR3	(Mandatory Inclusion) Organizations define specific, measurable phases. This creates the "Roadmap" used for all subsequent tracking.
UC4	Upload Impact Evidence	FR4, FR11	Organizations upload photos, invoices, and reports. Constraint: The system uses this to unlock subsequent phases as per FR11.
UC5	Generate Digital Receipt	FR5, FR6	(Automated) Triggered immediately after any donation or pledge. It uses transaction hashing to create a secure, unique ID for donor guarantee.
UC6	Manage Donor Dashboard	FR7, FR10	Provides donors with a view of their specific impact, showing real-time progress bars for cash, materials, and volunteers.
UC7	Donate & Pledge Items	FR8	Allows users to select specific material needs from the organization's list and commit to providing them.

Use Case ID	Use Case Name	Related FR	Description
UC8	Skill-Based Volunteer Matching	FR9	Uses professional tags (Medical, Teaching, etc.) to match volunteers with specific organizational task requirements.
UC9	Notify Supporters	FR12	(Automated) Sends push notifications/emails when milestones are reached or impact reports are uploaded.
UC10	View Transparency Ledger	FR13	A public-facing module allowing any user to audit the history of resource utilization across the platform.
UC11	Verify Volunteer Hours	FR14	The organization reviews and approves hours logged by volunteers to update the campaign success metrics.
UC12	Monitor & Flag Activity	FR15	Administrative dashboard use case to monitor organizational compliance and flag profiles missing evidence.
UC13	Geolocation Discovery	FR16	Integrates map-based searching to help donors find organizations in their immediate vicinity.
UC14	Generate End-of-Campaign Report	FR17	(Automated) Once all milestones are verified, the system generates a final summary for all contributors.
UC15	Export Audit Data	FR18	Allows organizations to download donor lists and contribution logs for tax and internal auditing.

2.6.2.2 Description of Use-Case Model

UC1: Register & Manage Org Profile

Field	Description
Actor	Charity Organization Representative
Pre-condition	The organization must have valid legal registration documents.
Main Flow	1. Actor selects "Register Organization." 2. Actor enters mission statement, contact details, and uploads legal registration/tax IDs. 3. System validates document format and saves the profile as "Pending Verification." 4. Admin verifies documents and activates the profile.
Ex. Flow	E1: Invalid Documents. If the uploaded file is corrupted or invalid, the system notifies the actor to re-upload.
Post-condition	The organization has a public-facing profile ready to launch campaigns.

UC2: Create Multi-Modal Campaign

Field	Description
Actor	Charity Organization Representative
Pre-condition	Profile must be "Active."
Main Flow	1. Actor initiates "New Campaign." 2. Actor defines title, story, and deadline. 3. Actor sets targets for: Financial (Cash), Material (Items), and Volunteers (Skills). 4. System triggers UC3 (Include) to set milestones. 5. System publishes the campaign.
Ex. Flow	E1: Missing Targets. If the actor leaves all targets at zero, the system prevents publication.
Post-condition	A campaign is visible to the public with a live progress bar.

UC3: Set Measurable Milestones (Included in UC2)

Field	Description
Actor	Charity Organization Representative
Pre-condition	Campaign details are entered.
Main Flow	1. Actor adds a Milestone (e.g., "Purchase 50 Wheelchairs"). 2. Actor defines the specific verification document required for this milestone. 3. System links these milestones to the campaign progress bar.
Ex. Flow	E1: Vague Milestone. If no measurable goal is entered, the system requests specific quantities.

UC4: Upload Impact Evidence

Field	Description
Actor	Charity Organization Representative
Pre-condition	A milestone has been reached in an active campaign.
Main Flow	1. Actor selects the specific milestone.

Field	Description
	2. Actor uploads "Evidence" (Site photos, signed delivery receipts, or invoices). 3. System timestamps the upload and notifies the Admin.
Ex. Flow	E1: Milestone Bypass. If the actor tries to update a later milestone without uploading evidence for the previous one, the system blocks the action (FR11).

UC5: Generate Secure Receipt (Automated)

Field	Description
Actor	System (triggered by Donor/Volunteer)
Pre-condition	Successful financial donation or material pledge verification.
Main Flow	1. System fetches donor data and organization legal data. 2. System generates a unique Transaction Hash ID. 3. System creates a PDF receipt.

Field	Description
	4. System emails the receipt to the donor and stores a copy in the Transparency Ledger.
Post-condition	Donor has a legally and technically verifiable proof of contribution.

UC7: Donate & Pledge Items

Field	Description
Actor	Donor
Pre-condition	An active campaign is selected.
Main Flow	1. Donor selects "Donate Cash" or "Pledge Material." 2. For materials, donor selects items from the organization's specific list. 3. For cash, donor completes payment via gateway. 4. System triggers UC5 (Include) for receipting.
Ex. Flow	E1: Payment Failure. If the gateway returns an error, the system cancels receipt generation and notifies the donor.

UC8: Volunteer Skill-Matching

Field	Description
Actor	Volunteer
Pre-condition	Volunteer has a profile with defined "Skill Tags."
Main Flow	1. Volunteer searches for tasks. 2. System filters tasks where Organizational Need Tags = Volunteer Skill Tags. 3. Volunteer applies for the match. 4. Organization approves the application.

UC11: Verify Volunteer Hours

Field	Description
Actor	Charity Organization Representative
Pre-condition	Volunteer has logged their hours.
Main Flow	1. Actor reviews the "Service Log" submitted by the volunteer.

Field	Description
	2. Actor clicks "Verify." 3. System updates the campaign's "Human Capital" progress percentage.
Ex. Flow	E1: Discrepancy. Actor can "Reject with Comment" if the hours do not match actual work done.

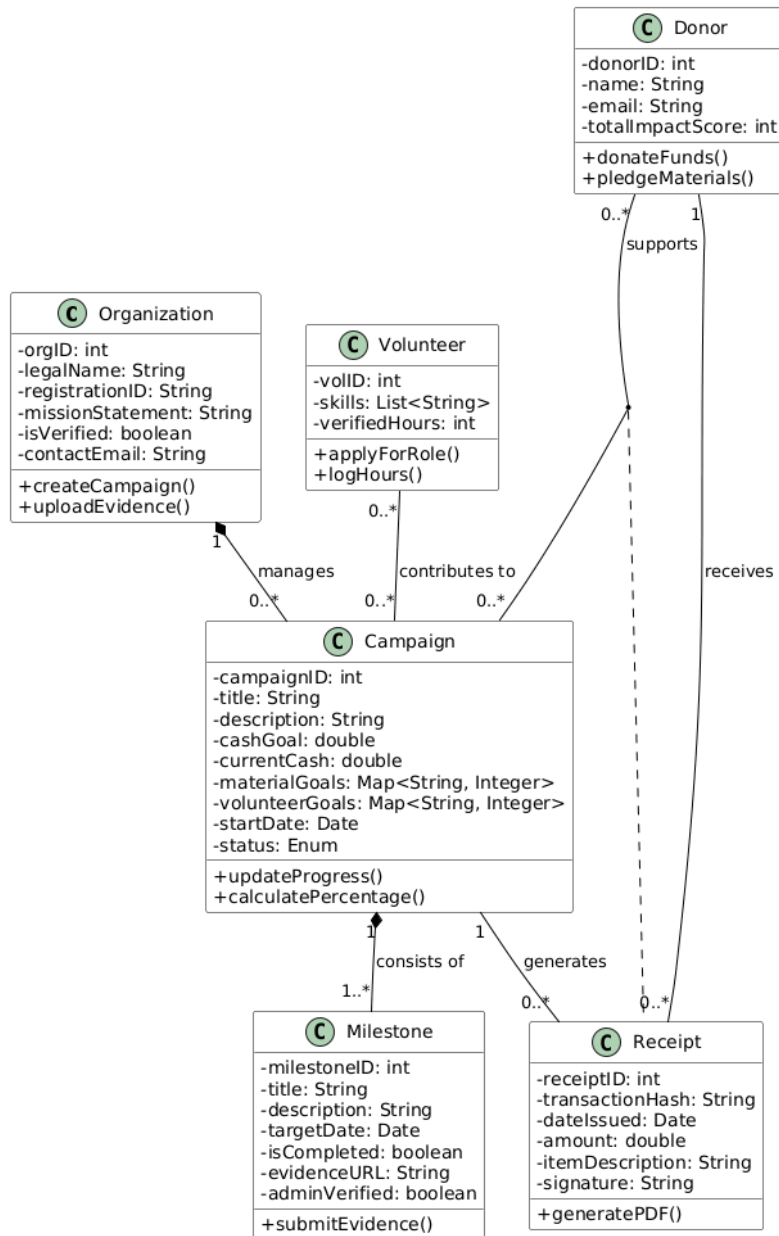
UC12: Monitor & Flag Activity

Field	Description
Actor	System Admin
Main Flow	1. Admin views the "Transparency Dashboard." 2. Admin reviews uploaded evidence (UC4). 3. If evidence is fraudulent or missing, Admin flags the profile. 4. System marks the organization with a "Warning" badge for public view.

UC15: Export Audit Data

Field	Description
Actor	Charity Organization Representative
Main Flow	1. Actor selects a date range and campaign. 2. System aggregates all donor names, transaction IDs, and values. 3. System generates a secure CSV/PDF file for download.

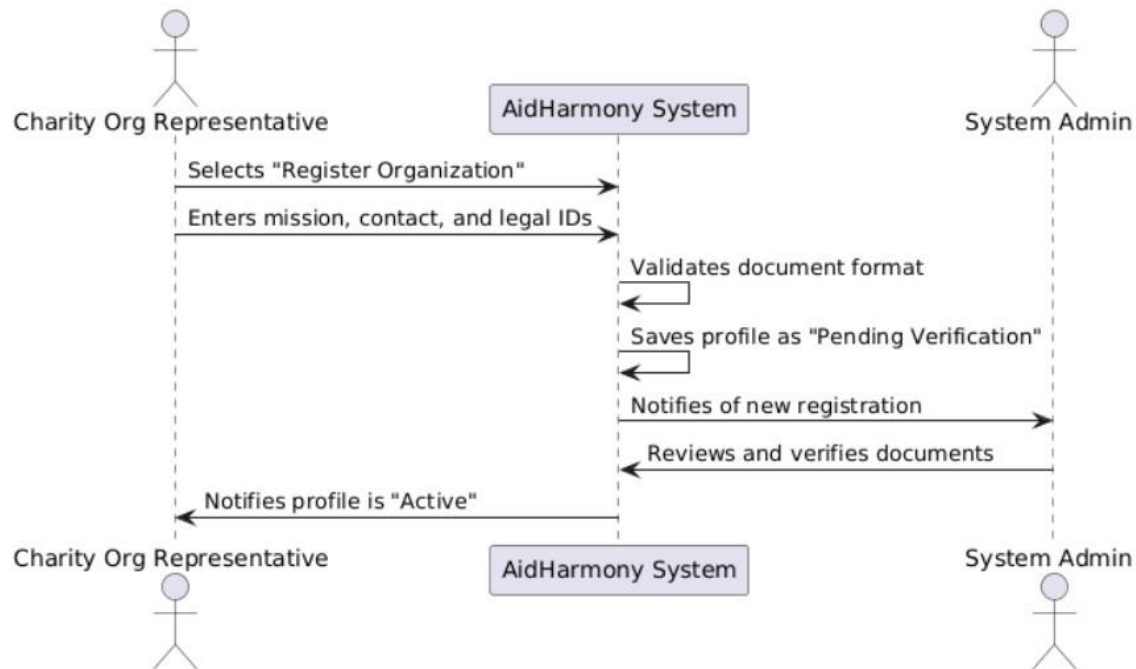
2.6.3.2 Class Modeling



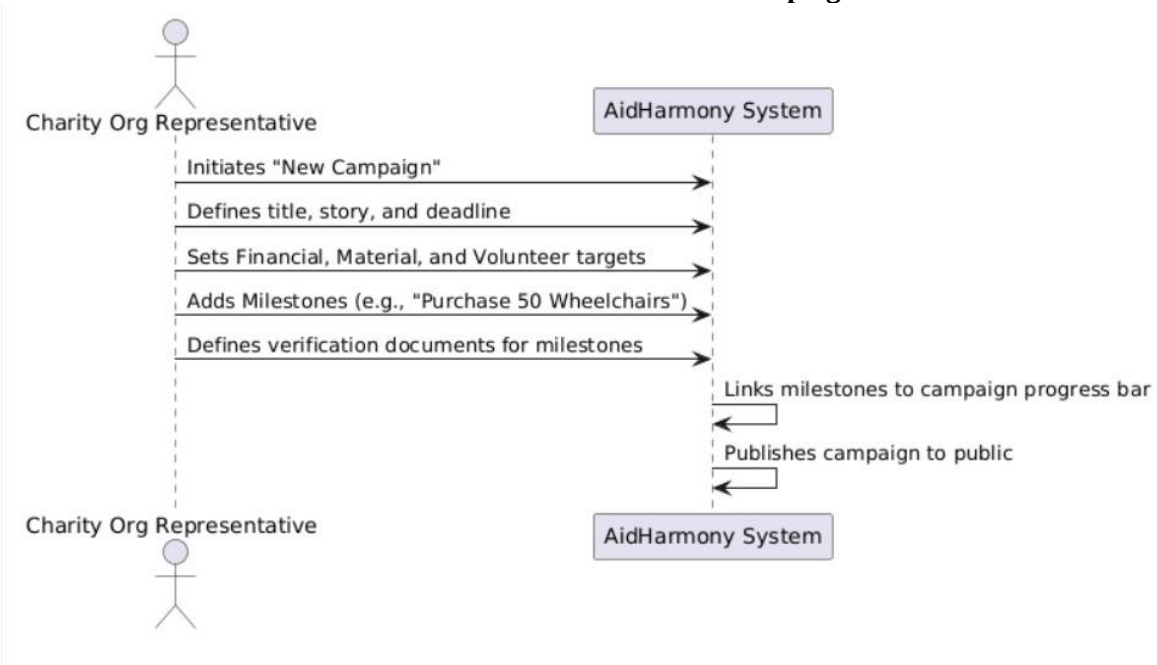
2.7 Dynamic model

2.7.1 Sequence Diagram

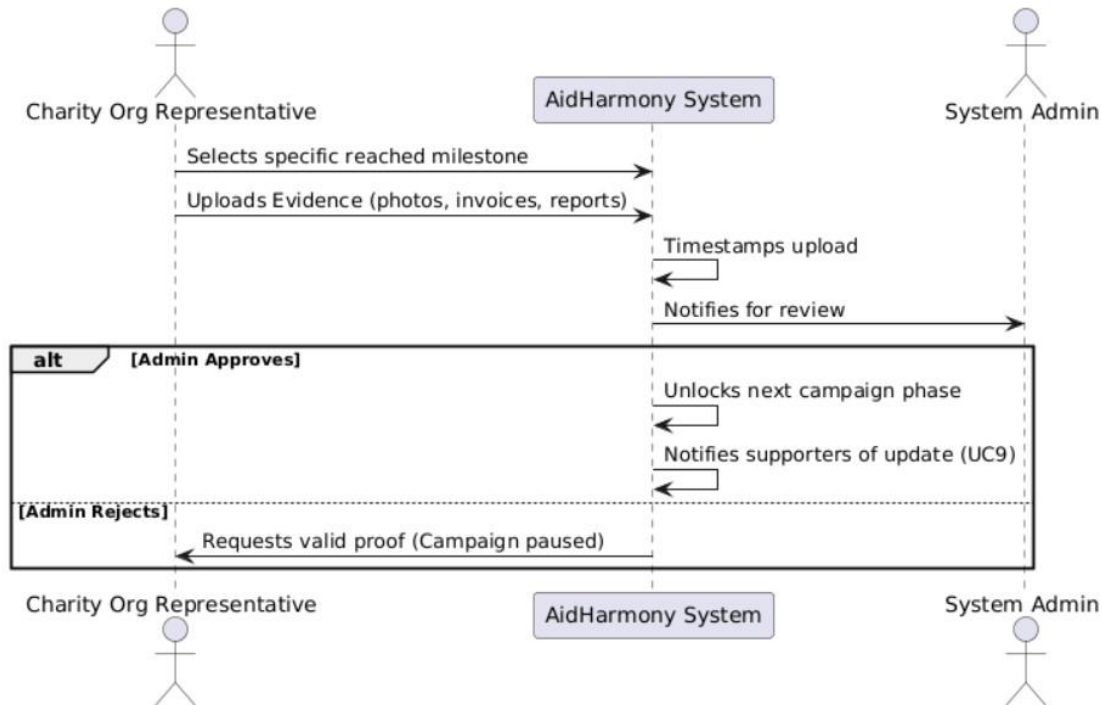
1. UC1: Register & Manage Organization Profile



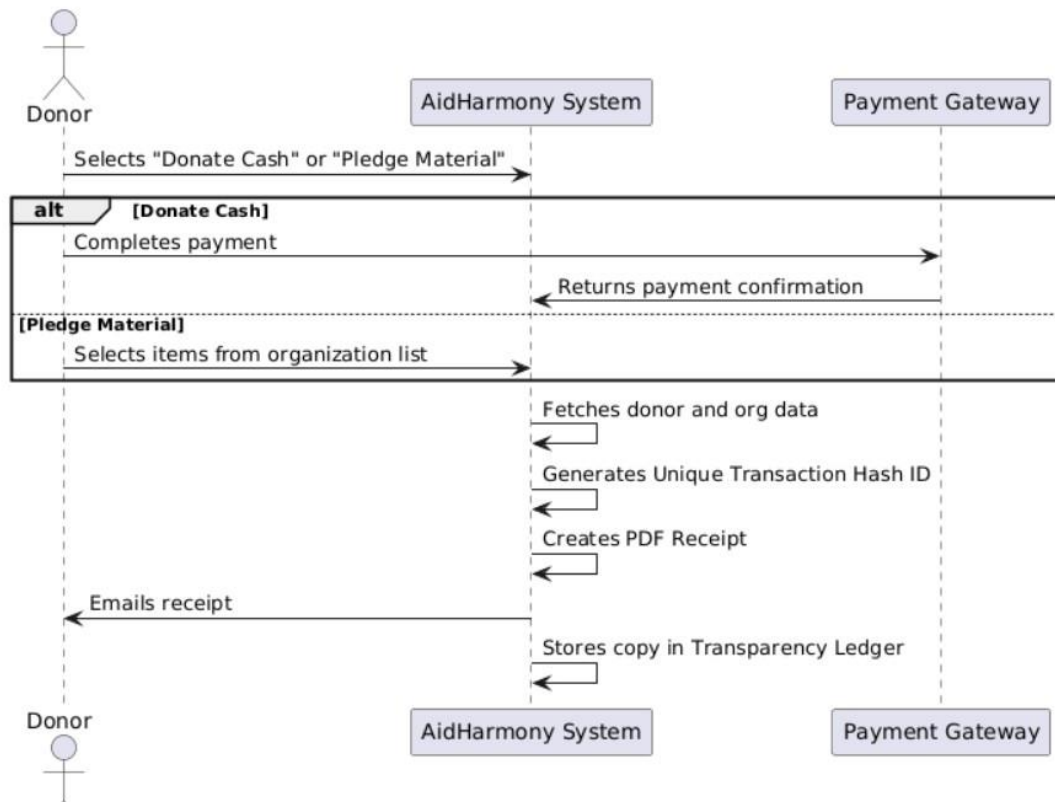
2. UC2 & UC3: Create Multi-Modal Campaign with Milestones



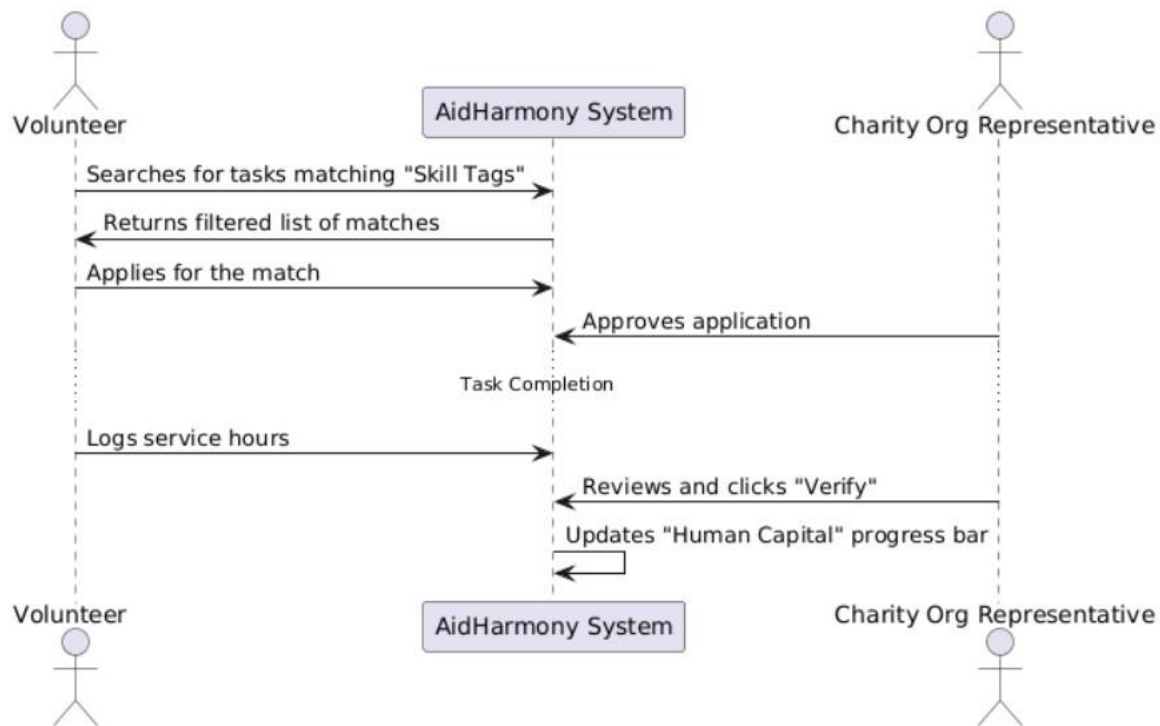
3.UC4: Upload Impact Evidence



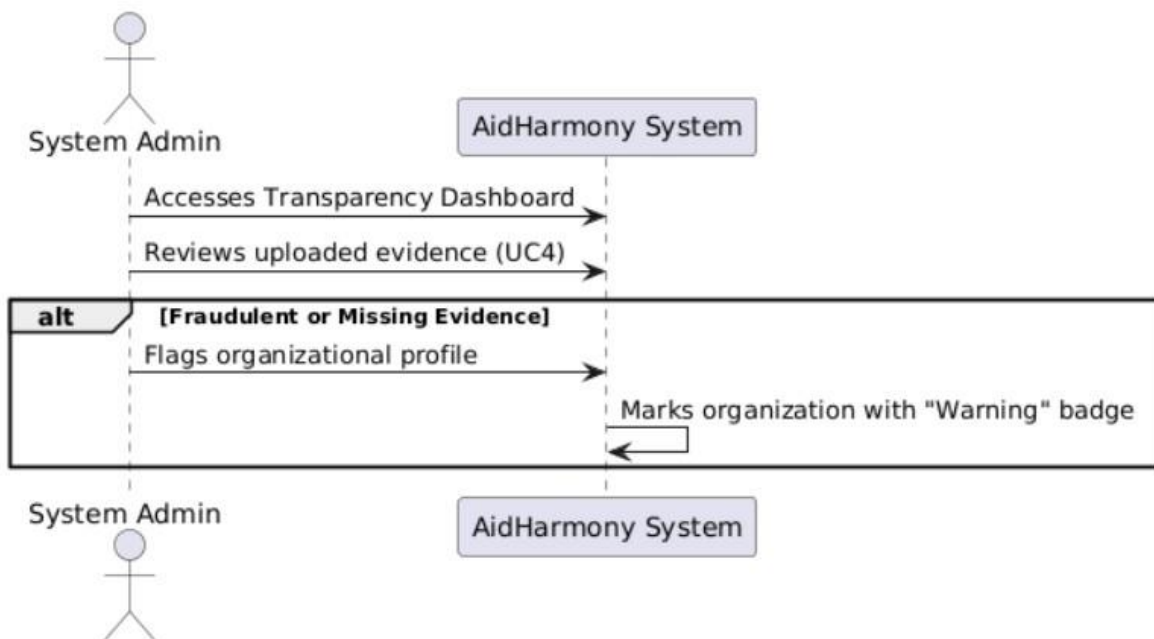
4.UC7 & UC5: Donate/Pledge & Generate Secure Receipt



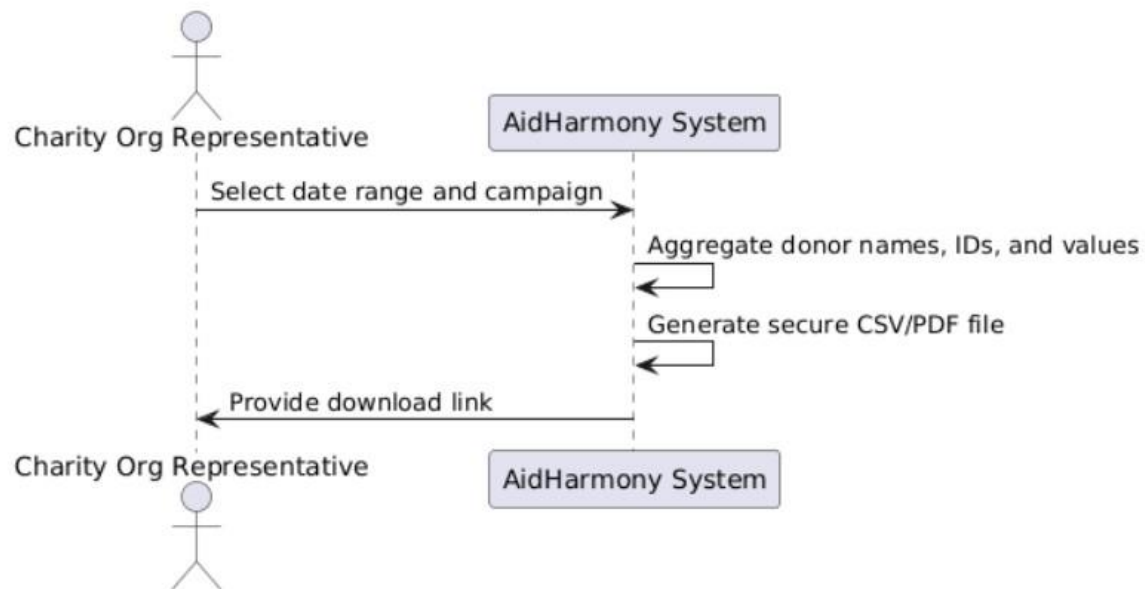
5.UC8 & UC11: Volunteer Skill-Matching & Verification



6.UC12: Monitor & Flag Activity



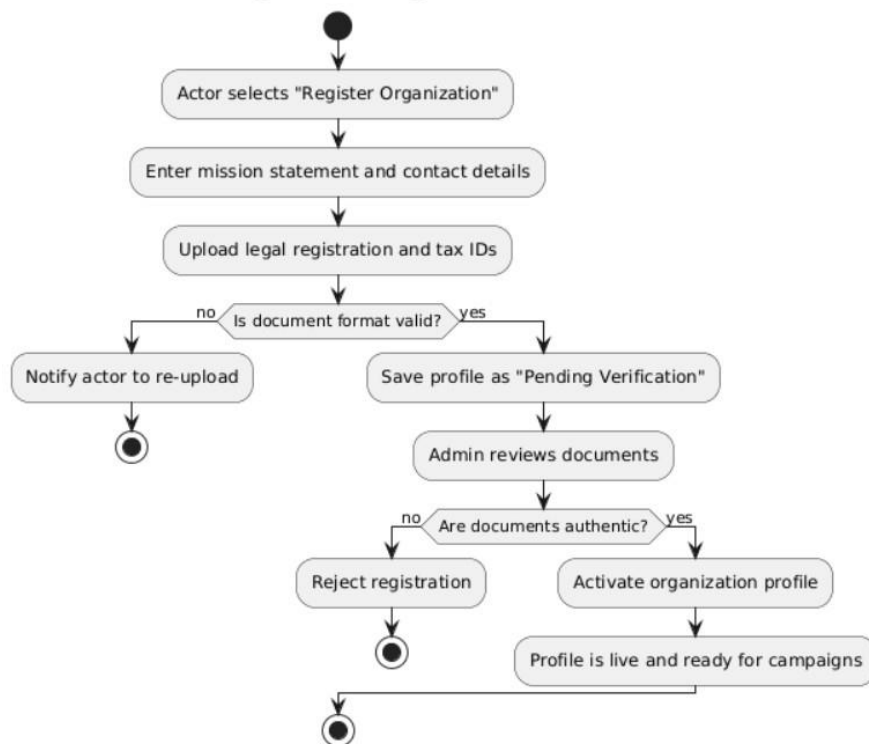
7.UC15:Export Audit Data



2.7.2 Flow diagrams

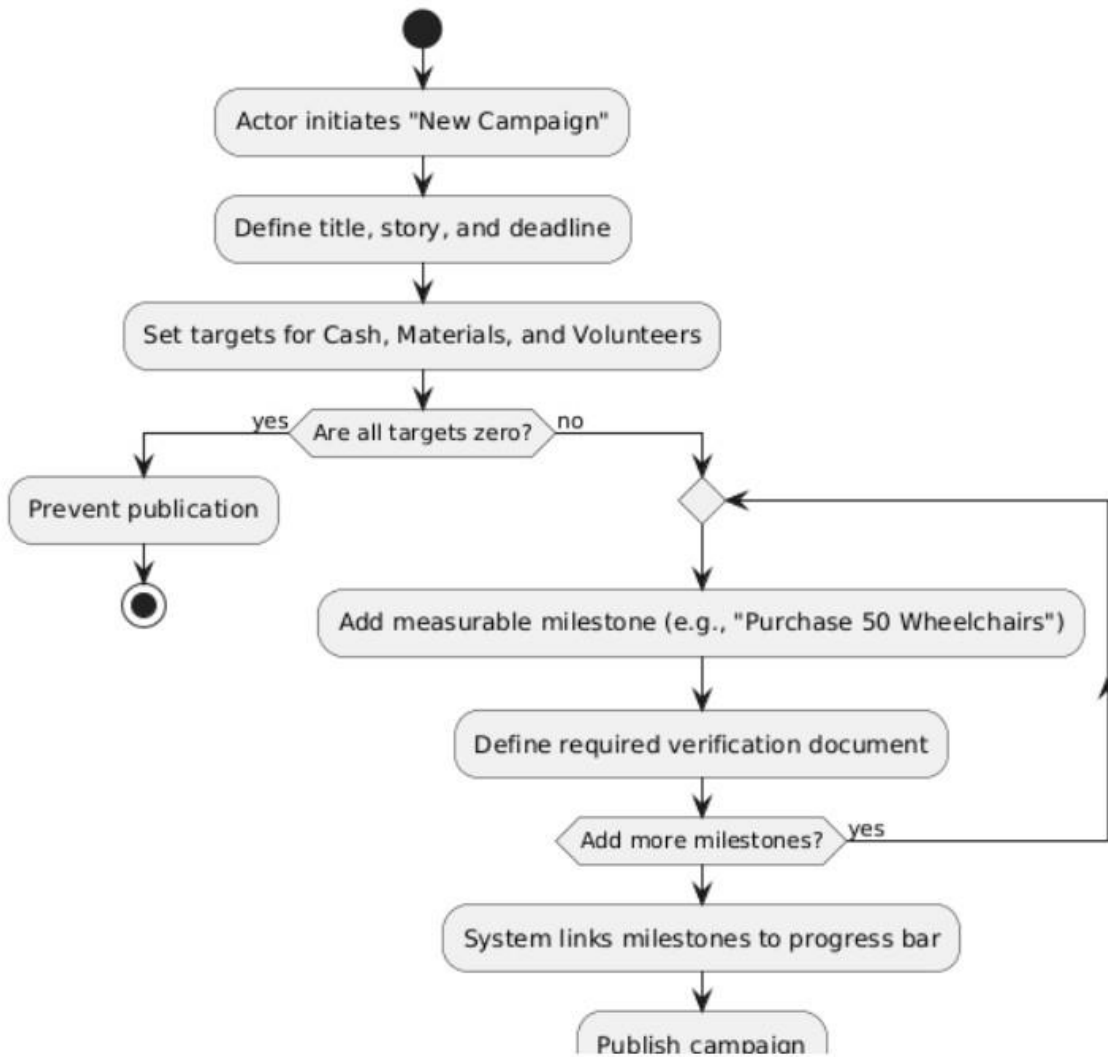
1. Organization Registration and Verification (UC1)

Organization Registration Flow (UC1)



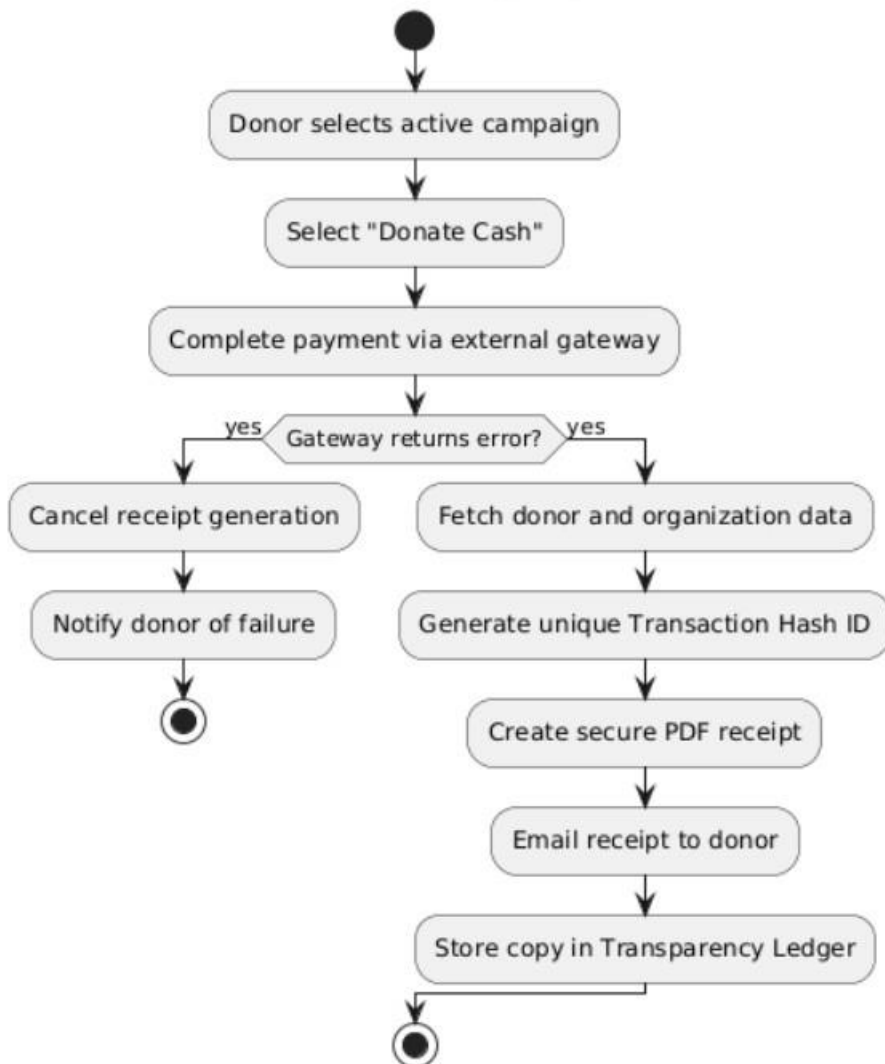
2. Multi-Modal Campaign Creation (UC2 & UC3)

Campaign Creation with Milestones (UC2 & UC3)



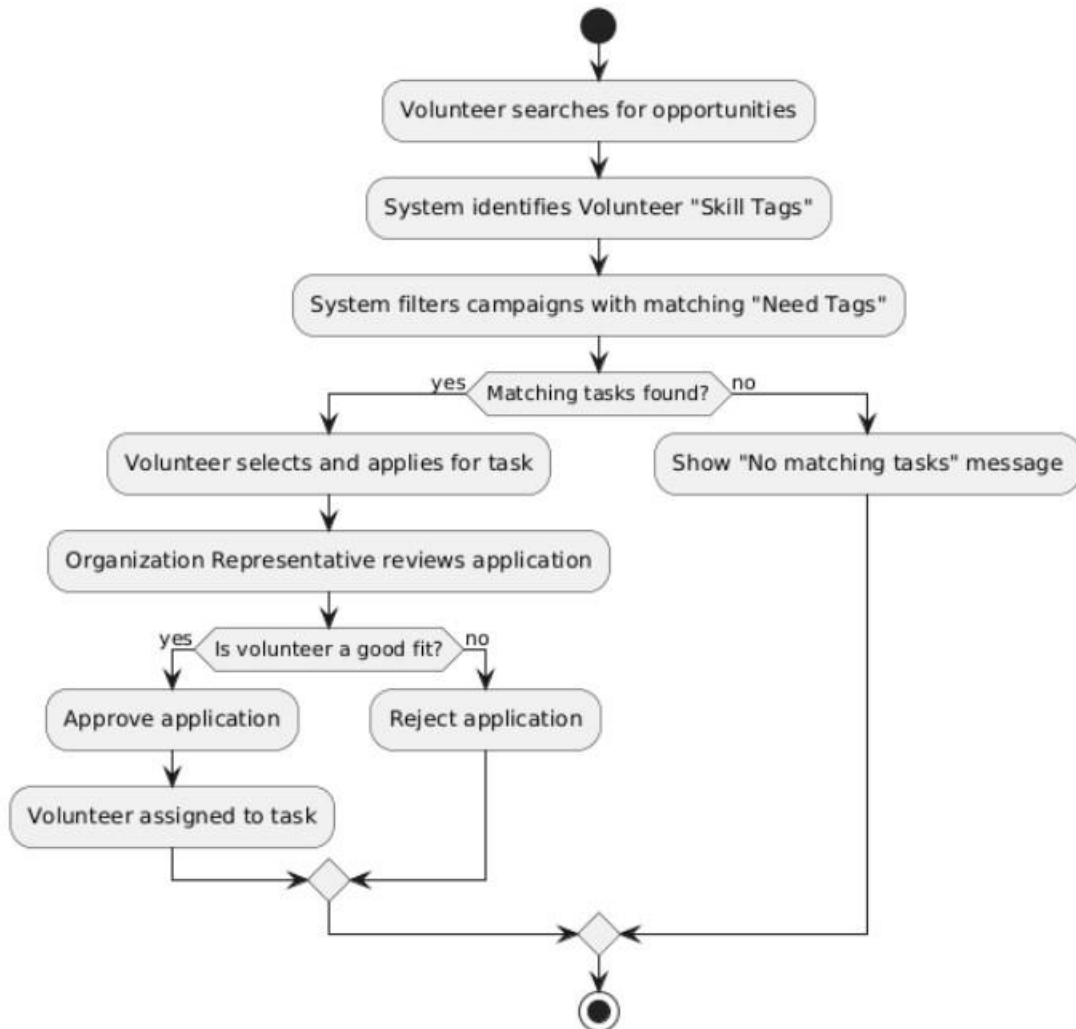
3. Financial Donation and Secure Receipting (UC7 & UC5)

Donation and Auto-Receipting (UC7 & UC5)

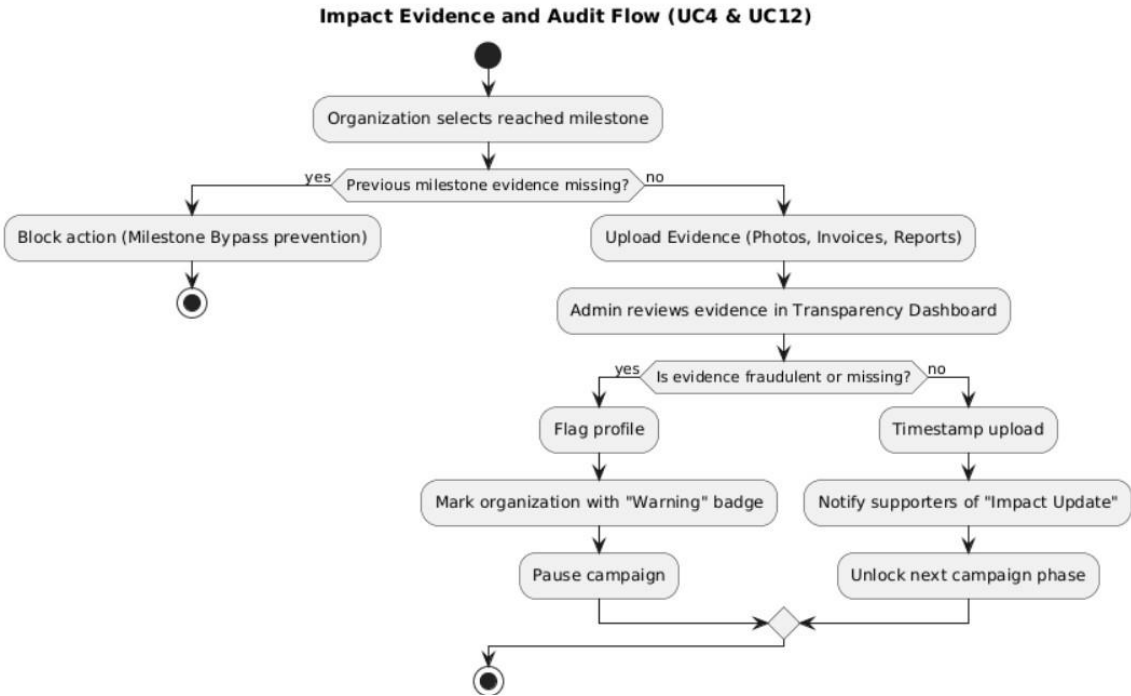


4. Volunteer Skill-Matching (UC 8)

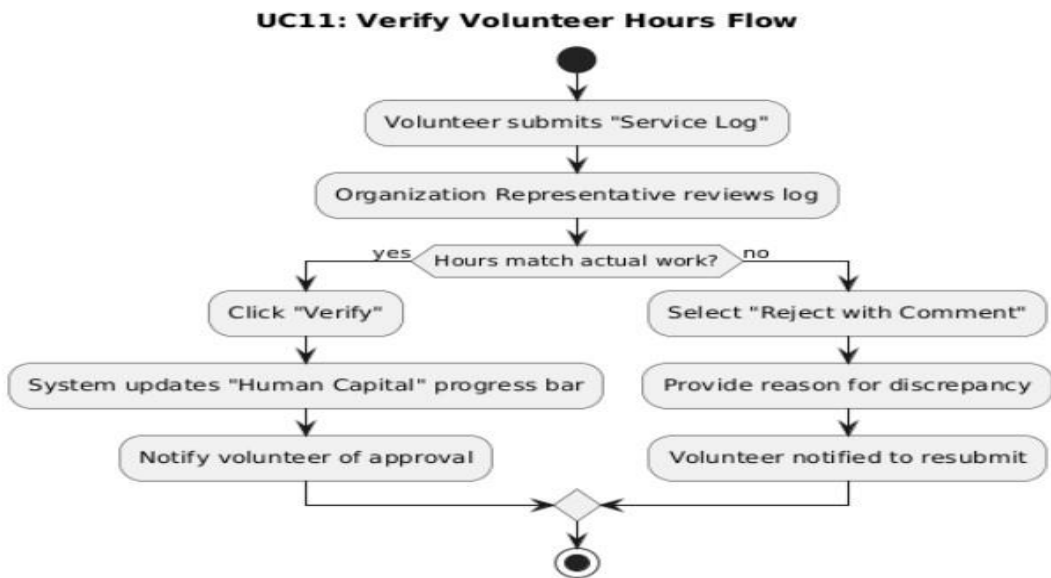
UC8: Volunteer Skill-Matching Flow



5. Milestone Reporting and Evidence Verification (UC4 & UC12)



6. UC11: Verify Volunteer Hours



7. Export Audit Data

UC15: Export Audit Data Flow

